“Let’s not ask our brains to be in two places at once. During lectures and class discussions, close your laptop and be fully present – we will open it again when it is time to code.”

PRESEMO Feedback

I like …

13:59 (28.10) » More Time on coding ☆

13:59 (28.10) » That we had to write code ourselves this time, encourages thinking about the code more. And liked that we had a bit more time for the coding ☆

13:59 (28.10) » That the AIS data is being worked on. It's something good for applications ☆

13:59 (28.10) » Time to code ☆

I wish …

14:11 (28.10) » Bring back the unzip notebook

13:59 (28.10) » All is good ☆

13:59 (28.10) » Everythings is ok ☆

Project report assessment criteria and presentation guideline is now available on the Mycourses page

Quiz Feedback

- All learning objectives were met – mostly "independently" and a few "with some support"
- Visualizing AIS data on a geographical map had one "not there yet" – we will revisit this today, ask questions during the exercise if something is unclear

Where we are on the course

We go at a pace that makes sense and adjust the depth/breadth of topics

Period I

- L1: Course orientation and foundations of marine data analysis
- L2: Exploratory Data Analysis (EDA)
- L3 & L4: Data pre-processing
- L5: From engineering questions to analytical models
- L6: Marine data analysis in practice

Period II

- L7: Regression modelling, evaluation, and refinement
- **L8 & L9: AIS data analytics**
- L10 & L11: Cluster analysis
- L12: AI-assisted qualitative data analysis
- Project demonstration

Aalto-yliopisto
Insinööritieteiden
korkeakoulu

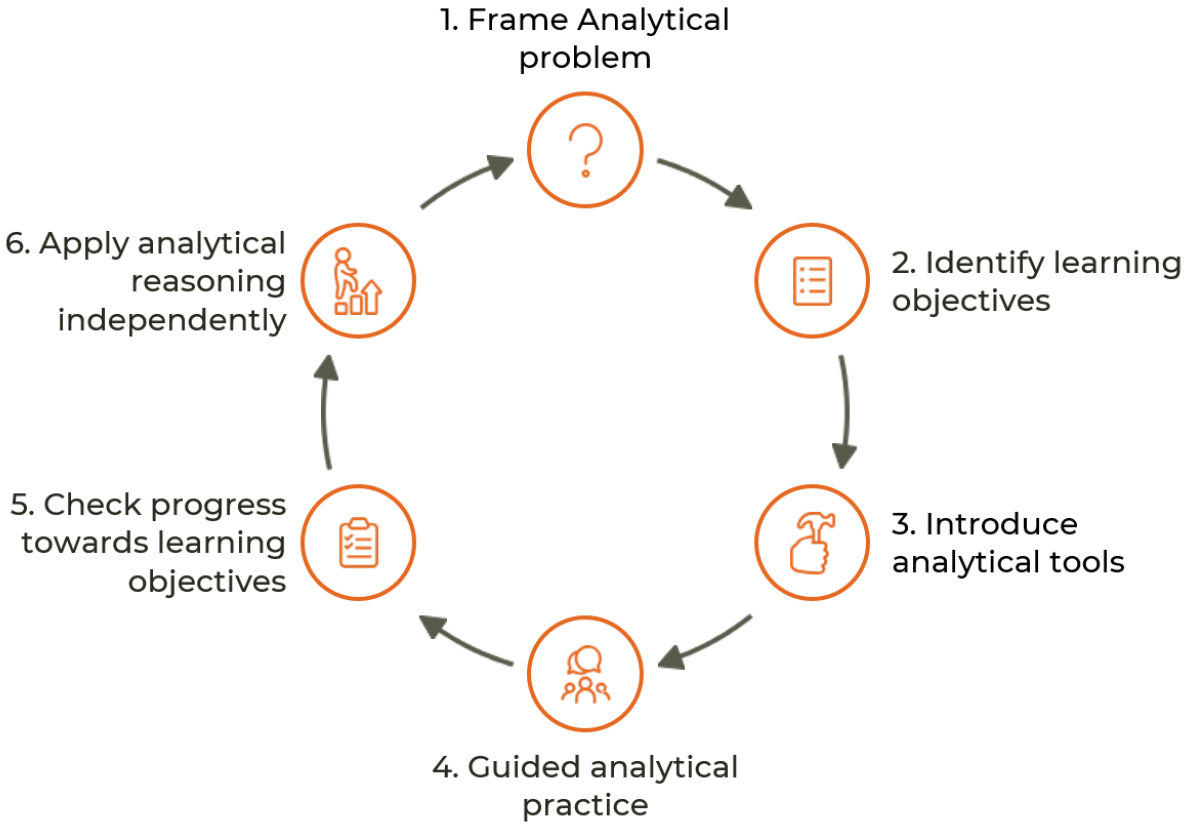
Introduction to AIS and AIS data analytics

MEC-E2016 Marine Data Analysis

Reference: Contents of this lecture are partially covered in

1. Guide to Maritime Informatics. Chapter 1 (Section 1.2), Chapter 4 (Sections 4.1 and 4.2).

Problem based learning cycle



Problem of the week (Due 11.11)

Grade 1 to 3: AIS data preprocessing

Apply appropriate preprocessing techniques to the AIS dataset, *AIS_data_timestamp.csv*, included in AIS_Preparation.zip. The ZIP file also contains a *libs* folder with Python functions for preprocessing and map visualization, and a *README.md* file explaining their use.

In your report, describe and justify each preprocessing action and explain its effect on data quality. Visualize the vessel positions before and after cleaning, compare the results, and discuss any remaining limitations that could affect further analysis.

Grade 4 to 5: Evaluation of AI-assisted preprocessing

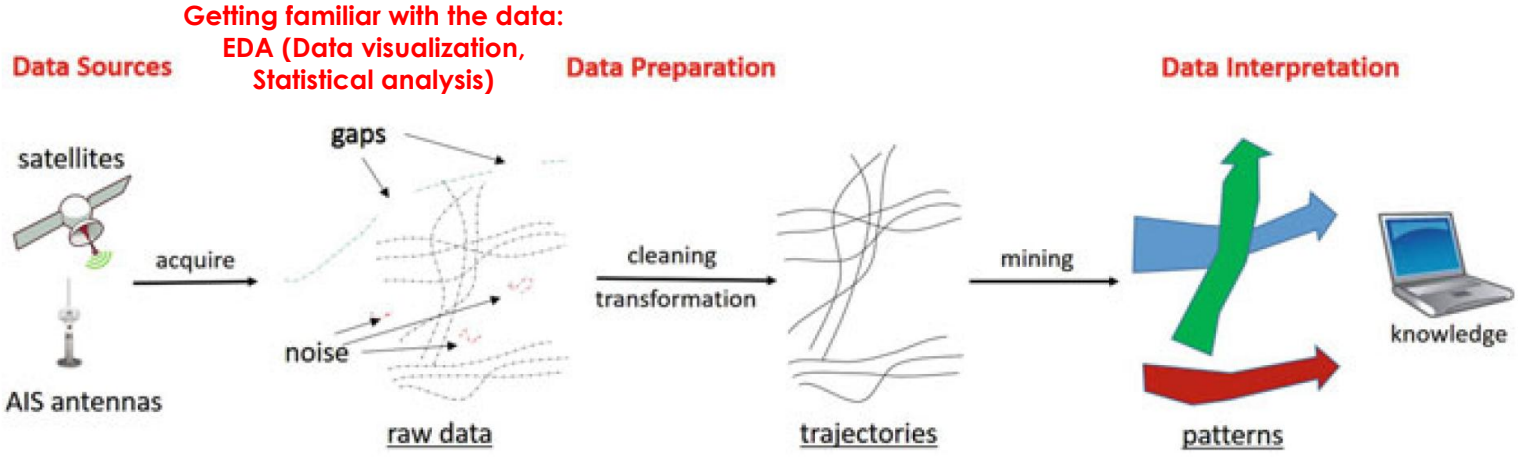
In addition to the above, repeat the AIS preprocessing task using a generative AI tool of your choice. State which tool you used and briefly describe how you instructed it. Compare its approach and results with your own work, focusing on the preprocessing decisions and their justification. Explain which AI suggestions you would accept, modify or reject, and support your assessment with specific examples from the dataset.

Learning objectives

At the end of this lecture, you should be able to

- Describe and perform (some of) the steps for AIS data analysis
  - Exploratory analysis of AIS data
  - Cleaning and transformation
  - Getting trajectories from location

Steps for AIS Data Analytics



Getting familiar with the data - Visualization

- The most natural step while exploring a dataset that involves movement of objects is to visualize the data on a map layer
- By doing this we can quickly
  - Get a general idea of the dataset under consideration
  - Identify obvious spatiotemporal patterns (e.g., errors in the reported location or timestamp)

Getting familiar with the data - Visualization

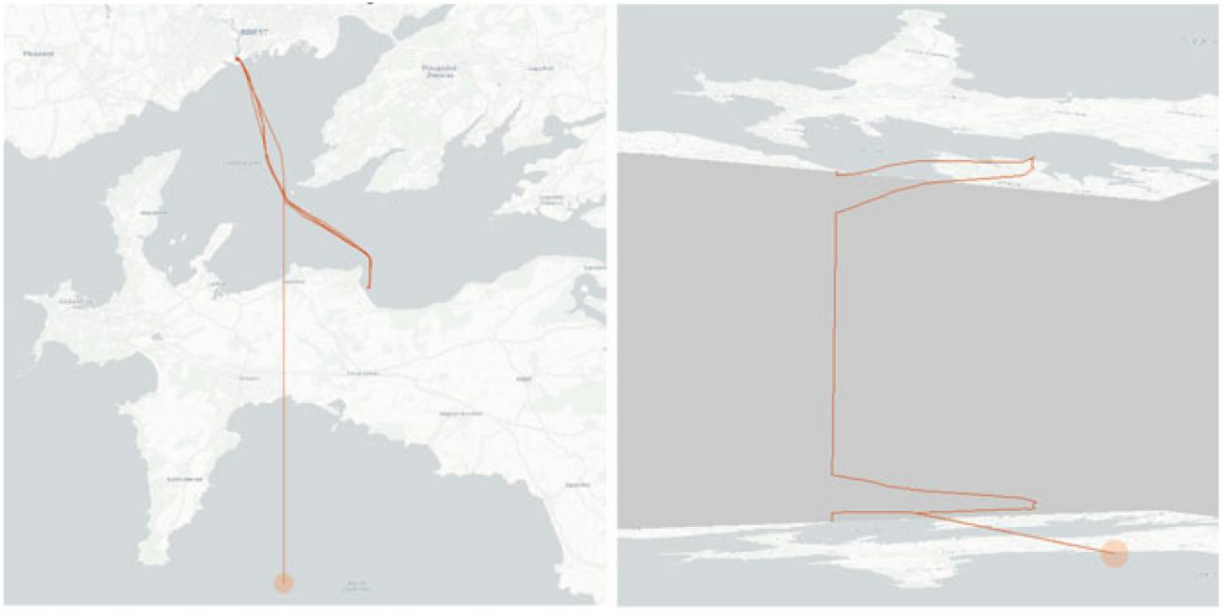


Fig. 4.3 An example trajectory with error in the reported location illustrated **a** in 2D and **b** in 3D (time as a third dimension in the z-axis)

Getting familiar with the data - Visualization

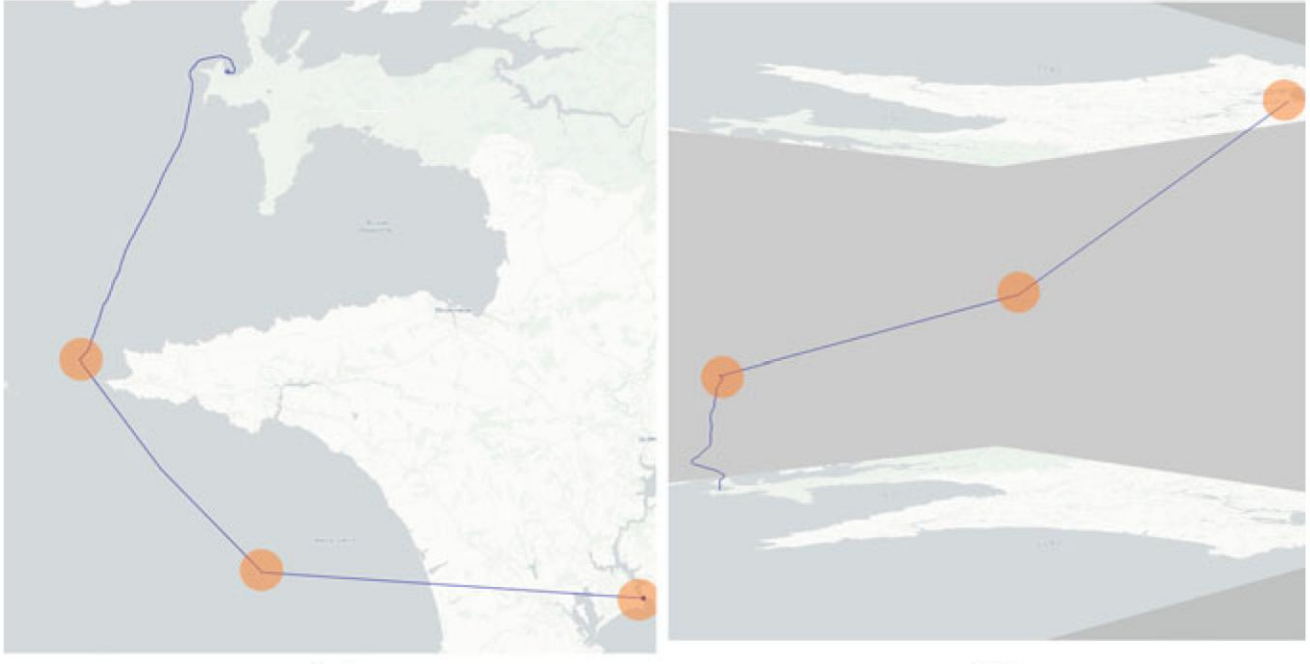
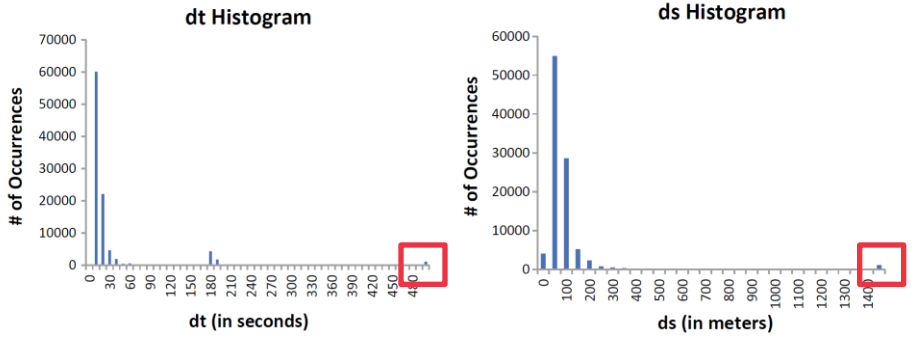


Fig. 4.4 An example trajectory with large temporal and spatial gap illustrated **a** in 2D and **b** in 3D (time as a third dimension in the z-axis)

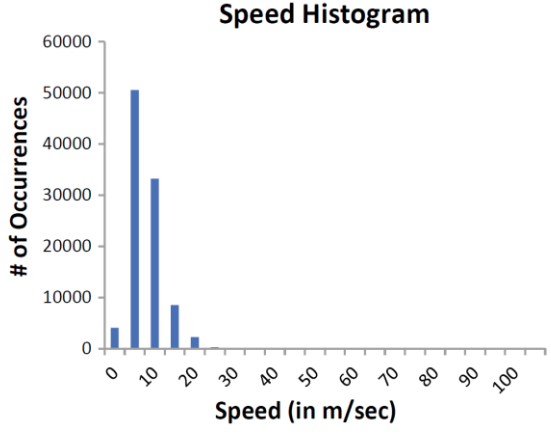
Getting familiar with the data – Statistical Analysis

- In order to gain real insight in a dataset, several statistics need to be calculated
- Example 1: Distribution of the temporal (dt) and/or spatial (ds) distance between two consecutive sampled positions of a vessel
  - These statistics can be useful to identify "unnatural" temporal and/or spatial gaps



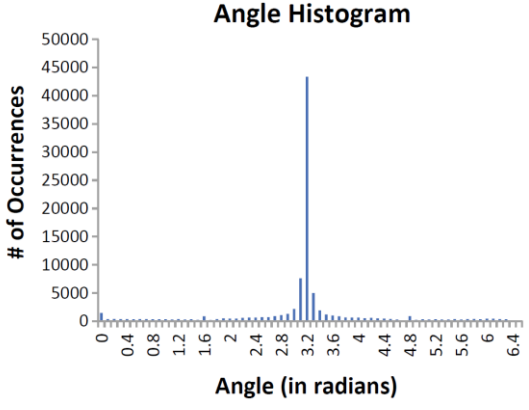
Getting familiar with the data – Statistical Analysis

- In order to gain real insight in a dataset, several statistics need to be calculated
- Example 2: Distribution of the speed between two consecutive samples of a vessel
  - This statistics can be useful to identify speeds that might be caused by erroneous situations (GPS/timestamping errors)



Getting familiar with the data – Statistical Analysis

- In order to gain real insight in a dataset, several statistics need to be calculated
- Example 3: Distribution of the angle made by three consecutive samples
  - This statistics can be useful to identify sharp turns that are not natural for maritime data and might be caused by errors.



- Majority of angles is between 2.4 and 5 rads (137◦–229◦) – as expected
- Sharp turns(e.g., lower than 90◦ or higher than 270◦) is a rare phenomenon

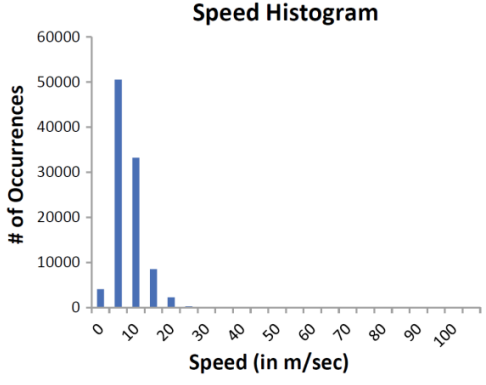
Data cleansing and transformation

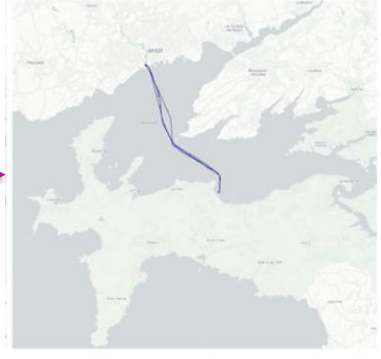
- Noise and errors are common issues when dealing with maritime data
- **Errors on the spatial position** can be categorized into two types
  - Systematic errors
    - Invalid GPS positioning readings, compared to the actual location of the vessel
    - Occur mainly due to the insufficient number of satellites in view
    - Approach to handle: Identify and remove
  - Random errors
    - Small distortions from the true values, up to a few meters
    - Caused by the satellite orbit, clock or receiver deficiency
    - Approach to handle: Decrease influence by smoothing

Systematic errors

- Approach to handle: Identify and remove
  - Design and implementation of automated "filters" which identify and remove "outliers".
  - Filter: Use the statistical analysis to set a threshold to decide whether a reported timestamped position should be considered as noise or not







Observation: "normal" speed between two consecutive positions reaches up to 30 m/s

Threshold: maximum allowed speed of a vessel, $V_{Max} = 30\ m/s$

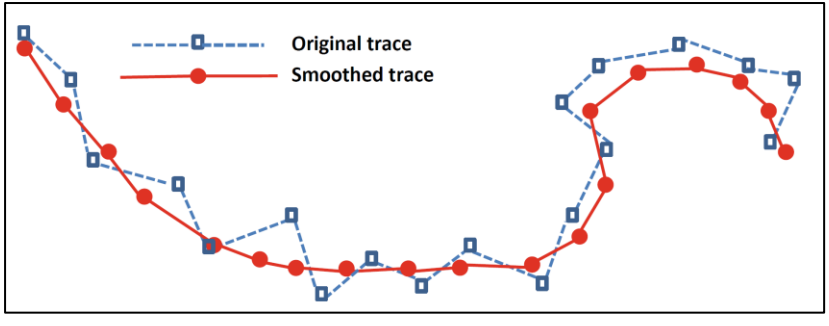
Outlier automatically removed as speed of the erroneous point was approximately 5000 m/s

Random errors

- Approach to handle: Decrease influence by smoothing
- Smoothing techniques
  - **Kernel-based smoothing**
  - Kalman filter

Random errors – Kernel based smoothing

- Kernel-based smoothing uses information from neighboring trajectory points to reduce local fluctuations and produce a smoother path.
  - **Choose the bandwidth:** This controls how far the smoothing extends around each position.
  - **Calculate a weighted average using the chosen bandwidth:** The kernel function (e.g., Gaussian, Epanechnikov, and uniform) determines how much each nearby observation contributes. Replace the original value with this weighted average.
  - **Repeat across the data:** Plot the new values to show a smooth curve.



- What are we averaging when smoothing a trajectory?
- Does "nearby" mean close in time or on the map?
- Is a smoother trajectory necessarily more accurate/better?

Yan, Z., Parent, C., Spaccapietra, S., Chakraborty, D.: A hybrid model and computing platform for spatio-semantic trajectories. In: Proceedings of ESWC, pp. 60–75 (2010)

Getting from locations to trajectories

- Input data of each vessel are long sequences of sampled positions
  - Their duration might even be equal to the duration of the whole dataset i.e. several months or years.
- To extract valuable knowledge from mobility data it is crucial to partition these long sequences into meaningful portions of movement (e.g., from port to port in the maritime domain)

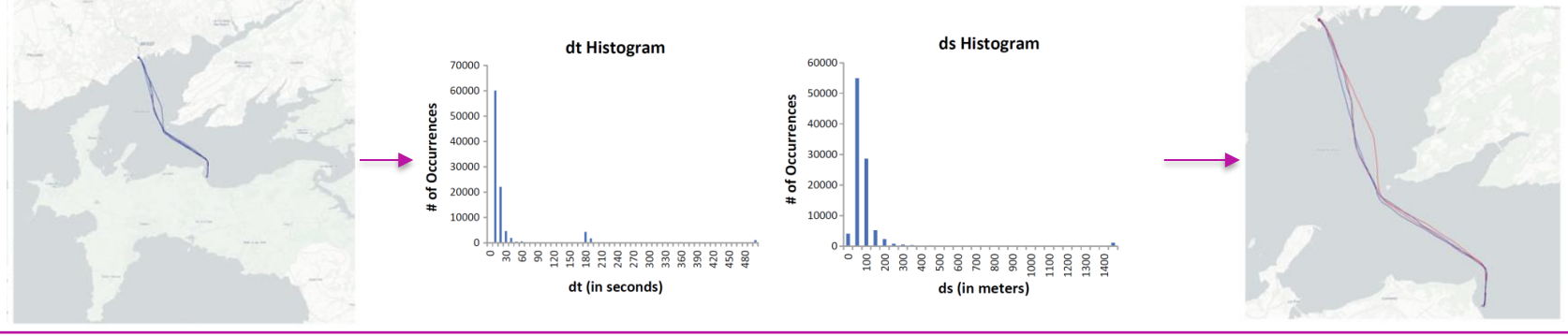
- Problem definition: define realistic identification rules in order to detect splitting points (p_i , t_i) that can divide the GPS feed into successive trajectories.

Getting from locations to trajectories

- Techniques for trajectory identification
  - Identification via raw gap
  - Identification by prior knowledge
  - Correlation-based identification
  - Identification via stop detection

Getting from locations to trajectories – Identification via raw gap

- A sequence of spatiotemporal points can be divided into trajectories with respect to the raw GPS gaps
- When the spatial distance between two consecutive timestamped positions exceeds a distance threshold G_{sp}
- OR when the temporal duration between two consecutive timestamped positions exceeds a temporal threshold G_t
- Use statistical analysis to set the threshold



Getting from locations to trajectories – Identification by prior knowledge

- Utilize some existing knowledge that might be useful in the trajectory identification procedure
  - Predefined time intervals - vessels performs scheduled routes during predefined intervals (e.g. weekdays from 8 a.m to 4 p.m)
  - Predefined spatial regions - vessels performs from port X to port Y

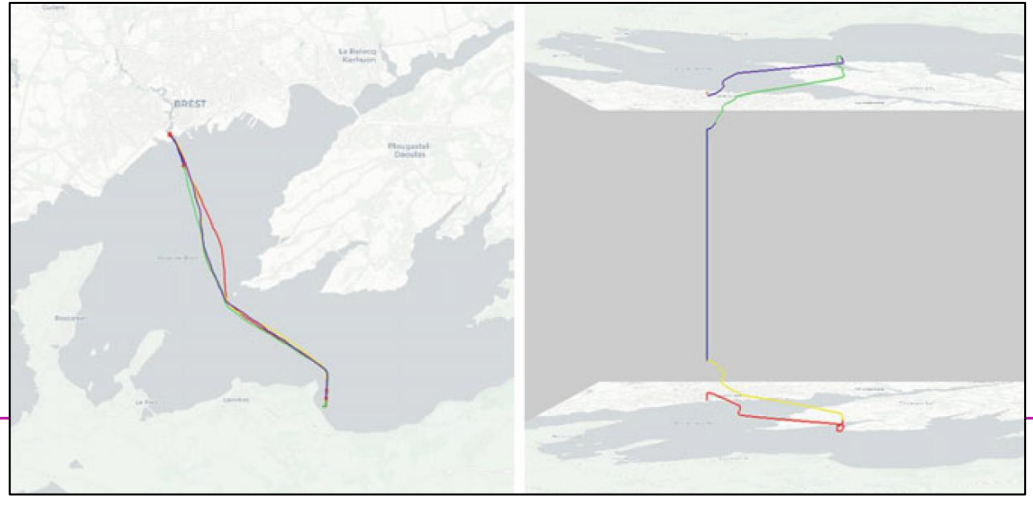
Getting from locations to trajectories – Correlation-based identification

- **Assume a trajectory in a transformed space:** instead of a time series of spatial positions, use a time series of relationships between positions, such as time intervals (dt), distances (ds), speed, or angle.

- **Apply time series segmentation techniques** to divide the trajectory based on the selected feature. For example:
  - Speed: Separate travelling periods from prolonged stops.
  - Angle: Identify sections with different directions of travel.
  - Time interval (dt): Separate sections divided by long gaps in observations.

Getting from locations to trajectories – Identification via stop detection

- Inspired by the notion that a trajectory is defined as the movement of an object between two "stops"
- problem of trajectory identification -> stop detection problem
- Given a sequence of spatiotemporal points of a vessel, identify portions of movement where the vessel has not moved (i.e., speed v is significantly low) for a duration of at least δt

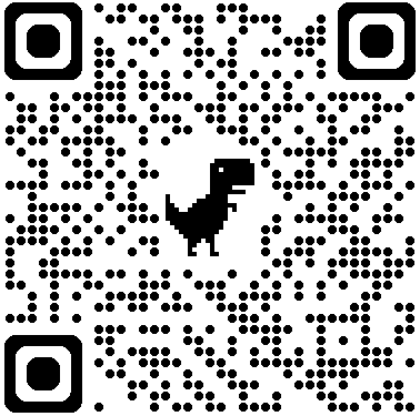


Exercise for today

- Find a peer and work through the Python exercise together.
- Download this week's **AIS_main_L9.ipynb** (available under this week's learning material) and put it in the **AIS_Preparation** folder from last week**.** Make sure it is ready to run.
- The code is already provided. You only need to enter the correct CSV file path so that the notebook can read the **AIS_data_timestamp.csv.**
- Work through the notebook together: **uncomment and run the code, examine the results, and answer the discussion questions**.
- We will discuss selected questions together as a class afterwards.

Quiz

- Complete your quiz here.



Feedback

- Provide your feedback here

